Pranav Joshi

Email: pranav.joshi@iitgn.ac.in | Phone: +91 8698955551 | GitHub | Linkedin

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PROFESSIONAL SUMMARY

Data engineer and analyst with experience in large-scale SQL systems, Retrieval-Augmented Generation (RAG) pipelines, statistical modeling, Machine Learning (ML) and time series data. Skilled in end-to-end development from data ingestion to AI integration, with a strong foundation in computer science and applied mathematics.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, SQL, GAP, C, Verilog, MATLAB
- **Proficient in** Pandas, NumPy, SciPy, Matplotlib, Flask, Django
- **Worked with** opencv, socket, Selenium, sqlite3, oracledb, openai, ollama,
- **Databases:** Oracle, MySQL, SQLite
- **Frontend / UI:** HTML5 Canvas, Streamlit, Jinja, StencilJS
- **Developer Tools & Platforms:** Docker, Git, SSH, phpMyAdmin, PowerShell, Bash
- **Concepts:** RAG systems, AI agents, statistical modeling, time series data, data visualisation, Machine learning, Feature Engineering, Hypothesis testing

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EDUCATION

B.Tech in Computer Science and Engineering, IIT Gandhinagar
CGPA: 9.59 out of 10
Dean's List in semesters 2, 4, 5, and 6
Relevant Courses: Computer Vision, Machine Learning, Data Science, Databases, Signals & Systems, Biostatistics, Compilers, Computer Networks

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PROFESSIONAL EXPERIENCE

Data Intern, Texas Instruments - May 2025 to July 2025

- **Text2SQL** - *June 2025 to July 2025*
  - Conducted an in-depth literature review of Text-to-SQL and Large Language Models (LLM) based database querying approaches.
  - Developed a metadata-free Retrieval-Augmented Generation (RAG) pipeline for column retrieval from enterprise SQL databases, handling **thousands of columns** and approximately **1 billion data points** without relying on metadata.
  - Designed a system to estimate the "importance" of database columns for improved retrieval relevance.
  - Built LLM-powered agents for table search, task decomposition, and SQL generation.
  - Created a Streamlit application with StencilJS-inspired components and Redux-style state management via OOP in Python.
  - Focused on scaling my Text2SQL application despite lack of metadata, which led to me developing a Jaccard Containment estimation technique suitable for large Oracle databases (see research section).
  - **Technologies:** Python, Oracle SQL, SQLite, LlamaIndex, OpenAI API, Streamlit, StencilJS
- **End-of-Life Report Automation**, - *May 2025 to June 2025*
  - Automated the analysis of electronic parts flagged for end-of-life status, replacing manual review and reducing time spent from multiple days to minutes.
  - Collaborated with business stakeholders to gather process requirements.
  - Read source code for SpringBoot projects to understand the working of API endpoints.
  - Reconstructed the same process using Oracle SQL queries.
  - Implemented a GUI for the automation script and packaged it as an executable
  - Solved many small issues of several employees related to SFTP and

Docker.

  - **Technologies:** Python, Pandas, Oracle DB, Docker, StencilJS, SpringBoot, SFTP

- **Awards and Achievements**
  - **PPO:** (Joining on July 13th) for the role of Data Engineer (not bond based).
  - Presented in one of Tech-Fridays on topics such as Pandas and Numpy

Individual Contractor for Industry-facing FPGA project

- **FPGA implementation of Total Focusing Method** *(MUSE lab, IITGN, currently working)*
  - Synthesised and Implemented a prototype on the Nexys 4 DDR board and later AC701
  - Interfaced the FPGA with PC using UART and Ethernet (RMII/RGMII physical layer)
  - Optimised by using dynamic programming, parallelism, pipelining, and streaming
  - Novel transducer domain partitioning and DDR3 usage allows for a frame rate of 30 fps for 128x128 pixel image using just one FPGA board.
  - Software Assisted Configuration allows for complex geometries, like usage of a wedge.
  - Created custom link layer protocol on top of Ethernet to reduce OS overhead.
  - Handed over the code to Modsonic Instruments Manufacturing Company Pvt. Ltd.
  - Consensus AI novelty report claims that the design is novel; future publication.

Research Internship, Computational Biology lab (CLOCK), IITGN, May-June 2026

- **SJL and Circadian Rhythm**
  - Did exploratory data analysis on physiological fine grain and day-granularity data provided by Ultrahuman
  - Applied data visualization techniques to identify two clusters in the data representing firmware changes

  - Used cosinor analysis with auto-regressive noise to capture the Circadian Rhythm

  - Defined new metrics for social jet lag (SJL) such as heart-rate-autocorrelation (HRA)

  - Characterised acute SJL events using changes in HRA
  - Trained transformer ANN for predicting HR, HRV and baselined against Cosinor Analysis
  - Paper writing is in progress and will likely lead to publication in next few months

Full stack Web-Developer

- **New Website for Indian Drought Monitor** *(WCL lab, IITGN, 25 May to 11 June 2026)*
  - Created an interactive map showcasing CDI/SPI as a heatmap, using HTML5 canvas

  - Created a local server for running LLM inference and a "Typst script to video" pipeline for automated summaries.

Academic Projects

P4 based SRv6 and INT implementation and testing *(Computer Networks, IITGN, Oct-Dec 2025)*

- Built a P4-based PDP, implementing IPv6 Segment Routing and SFANT style INT

- Wrote custom P4 parser and pipelines for BMv2 and integrated BMv2 with Mininet.

- Automated flow installation by running a shortest path algorithm on the controller.

- Benchmarked P4-based telemetry against SNMP/NetFlow.

ShuttleGo, a shuttle management system *(Databases, IITGN Jan-April 2025)*

- Created the MySQL+Flask application for a shuttle management system.
- Did a demonstration of the project once it was deployed using phpmyadmin.

AlgoLang, a custom PL, *(Compilers, IITGN Jan-April 2025)*

- Implemented the interpreter for a dynamically typed, lexically scoped, and extensible P.L. in Python.
- Implemented a compiler for a (large) subset of the PL that compiles to MIPS32.
- Created an end-to-end toolchain for the language (Compiler, Assembler, VM).
- Learnings: MIPS32, Recursive Descent Parsers, Basic PL theory

Assamese News Corpus (Hugging Face Dataset) *(IITGN, Jan-Apr 2025)*

- Extracted and curated a large-scale Assamese language corpus from online newspapers (*NeoNow*, *Dainik Barta*, *Assam NeoNow*) using Python, Selenium, Hugging Face API.

- Published the dataset on Hugging Face for public use in low-resource NLP research.

Person Velocity Detection CV Model *(IITGN, 2024)*

- Used YOLO-based object tracking and head detection models for person tracking.

- Used projective geometry and OpenCV to finally estimate the velocity

Open Source Contributions to GAP and SageMath *(IITGN, Jan-Jun 2024)*

- Collaborated with the author of a research paper on group theoretic algorithms.

- Reported bugs and implemented algorithms in GAP and SageMath.

Quantum Computing Enabled ISA *(Computer Architecture, IITGN 2024)*

- Read previous work such as quil and MIPS-Q as well as on QC, QML, QEC, etc.
- Developed a MIPS-like ISA that includes quantum instructions run using IBM's services.

Auto-Encode Neural Networks in System Verilog *(Digital Systems, IITGN, 2023)*

- Implemented an auto-encoder neural network on FPGA using system verilog

Classical Machine Learning *(ES335, IITGN, 2023)*

- Implemented classic ML models like decision trees, with visualisations and ensemble models like random forests and used boot-straping, K-fold split, etc.
- Read on optimisation methods like Momentum, SGD, batch gradient descent, Adam, etc.

Personal Projects

- Automatic differentiation system with backpropagation and neural network layers.

- Computational Physics Blog based on the course PH 509 at IITGN
- Trace; a slide based, programmable drawing application with stylus support, able to animate 3D plots
- Winner of HackRush'25 hackathon held at IITGN for the computer architecture track

Articles and Papers

Pranav Joshi, "Sampling-Based Estimation of Jaccard Containment and Similarity." *arXiv preprint, 2025.*

- Developed statistical models, Bayesian inference techniques, and bounds for set similarity estimation similar to the Lincoln-Peterson population size estimator.

- Empirically validated the models using simulated data.

- Gave guidance of applying this method for large databases with a rate/batch limit for querying.

- Applications: embedding propagation for RAG pipelines, plagiarism detection, literature search.